Raphaël Barboni

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Academic Positions

2025 – present **Postdoctoral fellow**, *Bocconi University, Institute for Data Science and Analytics*, Milan
mentored by **G. Zanella**.

Education

Fall 2025 **Ph.D. defense**, *École Normale Supérieure – PSL, Department of Mathematics*, Paris
Thesis: "Training deep neural networks: mathematical models and convergence results"
Referees: Mikaela Iacobelli (ETH Zürich), Joan Bruna (NYU)
Jury: Claire Boyer (Université Paris-Saclay), Borjan Geshkovski (Inria), Eric Vanden-Eijnden (NYU)

2022 – 2025 **Ph.D. in Mathematics**, *École Normale Supérieure – PSL, Department of Mathematics*, Paris
supervised by **G. Peyré** and **F.-X. Vialard**.

2018 – 2022 **Élève fonctionnaire**, *École Normale Supérieure – PSL, Department of Mathematics*, Paris

2020 – 2021 **M.Sc. Mathematics for Learning and Vision (MVA)**, *ÉNS Paris-Saclay*, with honors
Thesis: *"Convergence properties of Gradient Descent in the training of Deep Residual Networks"*,
supervised by **G. Peyré** and **F.-X. Vialard**.

2018 – 2019 **B.Sc. in Mathematics**, *École Normale Supérieure – PSL*, Paris, with honors
Thesis: *"Mean curvature flow, an introduction to geometrical flows"*,
supervised by **T. Ozuch**.

2016–2018 **"Classes préparatoires"**, *Lycée Henri IV*, Paris
Mathematics, Physics, Computer Science (MPSI-MP*).

Experiences

2022 **Research internship**, *ETH, Mathematics Department*, Zürich
"Statistical to computational gaps in Tensor PCA", supervised by **A. Bandeira**.

2020 **Research internship**, *Center for Mathematical Modeling (CMM)*, Santiago, Chile
"Hydrodynamical models for red tide phenomena in Quellón's bay", supervised by **C. Conca**.

Teaching

2023 – 2025 **Teaching assistant**, *Université Paris Sciences et Lettres (PSL)*, Paris
"Cycle Pluridisciplinaire d'Études Supérieures" (CPES), Mathematics, 2nd year.

2021 – 2023 **Teaching assistant**, *Université Paris Sciences et Lettres (PSL)*, Paris
"Cycle Pluridisciplinaire d'Études Supérieures" (CPES), Mathematics, 1st year.

2018 – 2022 **Oral examinations**, *Lycée Henri IV*, Paris

Awards

2022 **Ph.D. fellowship**, *PSL Graduate Programme in Mathematics and Applications*, Paris

Publications

Published

R. B., Gabriel Peyré, and François-Xavier Vialard. "Understanding the training of infinitely deep and wide ResNets with conditional optimal transport". In: *Communications on Pure and Applied Mathematics* (2025).

R. B., Gabriel Peyré, and François-Xavier Vialard. "On global convergence of ResNets: From finite to infinite width using linear parameterization". In: *Advances in Neural Information Processing Systems* 35 (2022).

Preprints

R. B., Sebastiano Grazzi, and Giacomo Zanella. "Hamiltonian dynamics for sampling on discrete spaces". In: *arXiv preprint arXiv:2608.17961* (2026).

R. B., Maarten V. de Hoop, Takashi Furuya, and Gabriel Peyré. "Training Infinitely Deep and Wide Transformers". In: *arXiv preprint arXiv:2605.17660* (2026).

R. B., Gabriel Peyré, and François-Xavier Vialard. "Ultra-fast feature learning for the training of two-layer neural networks in the two-timescale regime". In: *arXiv preprint arXiv:2504.18208* (2025).

Conferences

Oral presentations

- SIAM – Mathematics of Data Science, November 2026, Salt Lake City, United States
- International Conference on Curves and Surfaces, June 2026, Saint-Malo, France
- Journées SMAI-MODE, March 2026, Nice, France
- International Workshop on Operator Theory (IWOTA), July 2025, Twente, Netherlands
- Signal, Image, Geometry, Modeling and Approximation (SIGMA), June 2024, Luminy, France
- PDE Methods in Machine Learning (Banff IRS event), June 2024, Granada, Spain

Poster presentations

- Foundations of Computational Mathematics (FoCM), July 2026, Vienna, Austria
- Scalable MCMC Sampling, June 2026, ETH Zürich, Switzerland
- Physics of AI algorithms, January 2025, Les Houches, France
- Learning and Optimization in Luminy (LOL), June 2024, Luminy, France
- Workshop on Optimal Transport, from Theory to Applications, March 2024, Berlin, Germany
- Neural Information Processing Systems (NeurIPS), November 2022, New Orleans, USA
- International Conference on Curves and Surfaces, June 2022, Arcachon, France

Community

Reviewer

I reviewed for the following venues:

- Communications on Pure and Applied Mathematics,
- Annales de l'Institut Henri Poincaré, IMS journal,
- Applied and Computational Harmonic Analysis,
- Mathematics of Computation, AMS journal,
- ESAIM: Control, Optimisation and Calculus of Variations,
- Journal of Machine Learning Research,
- Machine learning conferences: Neurips, ICML, AISTATS.